



Commentary

Explosive type of contractions should not be avoided during resistance training in elderly

Eduardo Lusa Cadore^a, Ronei Silveira Pinto^a, Álvaro Reischak-Oliveira^a, Mikel Izquierdo^{b,*}^a Exercise Research Laboratory, Physical Education School, Universidade Federal do Rio Grande do Sul, Porto Alegre, RS, Brazil^b Department of Health Sciences, Public University of Navarra, CIBER de Fragilidad y Envejecimiento Saludable (CB16/10/00315), Tudela, Navarra, Spain

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This letter is a commentary regarding the Mini review by Fisher et al. (2017), entitled “A minimal dose approach to resistance training for the older adult; the prophylactic for aging”, which was recently published in the *Experimental Gerontology*, 99, 80–86. Although we recognize the experience of the authors in the resistance training research field, as well as we agree with the main message of the article, that is, a minimal dose of resistance training provides several health benefits in elderly individuals, we would like to complement some provided information, and argue, based on strong scientific evidence, against some affirmations.

The authors have mentioned some relevant studies which found similar strength and muscle mass gains when comparing one vs. multiple sets in older adults (Cannon and Marino, 2010; Abrahin et al., 2014; Galvão and Taaffe, 2005; Radaelli et al., 2014). While we agree that low volume of resistance training (RT) is a very efficient intervention because it provides marked neuromuscular and functional gains in elderly populations (Abrahin et al., 2014; Radaelli et al., 2014), especially in the short-term (i.e. ≤ 6–12 weeks), it cannot be unconsidered that there are some meta-analyses that showed advantage of two and three sets per exercise when compared to one set per exercise in older adults (Peterson et al., 2010; Borde et al., 2015). In addition, the mentioned study of Radaelli et al. (2014) observed similar magnitude of neuromuscular adaptations induced by one and three sets per exercise only until 13 weeks. After 20 weeks, these authors observed greater lower-body 1 RM, quadriceps muscle thickness, and muscle quality gains in the group of elderly women who performed three sets per exercise (Radaelli et al., 2014). Again, we agree that low volume is a very efficient dose of RT, but it should be mentioned based on the literature that greater volumes may be necessary to optimize neuromuscular adaptations in the long-term, especially in lower limbs.

The authors mentioned that “...the exercise protocols achieved more favorable adaptations where participants were required to lift a moderate to

heavy load (e.g. 50–90% 1-repetition maximum; RM) and either exercise to momentary failure (e.g. where they could not complete another repetition despite attempting to do so) or complete multiple sets of each exercise permitting cumulative fatigue and higher effort in later sets”. While maximal strength gains may be optimized when elderly perform RT using loads ranging from 70 to 79% 1 RM (Borde et al., 2015), or > 75% of 1 RM (Steib et al., 2010), we disagree that fatigue is a necessary stimulus to provide optimal neuromuscular adaptations. Although repetitions until concentric failure is often used in the RT prescription in elderly, inducing marked neuromuscular adaptations (Cadore et al., 2012; Radaelli et al., 2014), there is no evidence that this method of RT prescription provides additional adaptations when compared with submaximal repetitions in elderly populations. Indeed, in young populations, there are several studies which showed no difference between repetitions to failure or not to failure in the magnitude of maximal strength gains (Folland et al., 2002; Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Sampson and Groeller, 2016; Martorelli et al., 2017), muscle size gains (Sampson and Groeller, 2016; Martorelli et al., 2017), and maximal power (Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010). A fewer number of studies have shown some advantage on strength gains performing repetitions to failure (Rooney et al., 1994; Drinkwater et al., 2005), and these studies investigated young subjects. Along with no evidence of additional neuromuscular adaptations in elderly, performing multiple sets is associated with cumulative fatigue and higher effort in the later sets (i.e., induced by repetitions to failure), as mentioned by authors. Longer sets may elicit greater increases in the systolic and diastolic blood pressure, heart rate, as well as rate-pressure product. This effect is not necessarily explained by the fatigue, but because longer sets induce greater cardiovascular responses (Lamotte et al., 2005; Nery et al., 2010; Lovell et al., 2011; Gjovaag et al., 2015; Gjovaag et al., 2016). Therefore, this type of RT prescription should be avoided in elderly, especially those

* Corresponding author at: Department of Health Sciences, Public University of Navarra (Navarra), Campus of Tudela, Av. de Tarazona s/n. 31500, Tudela, Navarra, Spain.
E-mail address: mikel.izquierdo@unavarra.es (M. Izquierdo).

with hypertension or another cardiovascular disease.

Besides to discuss the information above mentioned by the authors, our main concern regarding the content of authors' review is the statement that "...trainees might performed controlled 2–4 s concentric: 2–4 s eccentric phases, which should maintain muscular tension throughout the range of motion, decrease momentum and reduce high forces by preventing explosive movements.". This statement is against a large and solid body of knowledge on the literature regarding RT prescription in elderly. The prescription of explosive muscle actions during RT, known as explosive RT, high-speed/velocity RT or power training, and its benefits in the neuromuscular function in elderly had been previously demonstrated by classical studies (Häkkinen et al., 1998; Skelton et al., 1995; Jozsi et al., 1999; Izquierdo et al., 2001).

There are important arguments to justify the inclusion of explosive type of RT in elderly. First, muscle power output and rate of force development are strongly associated with the capacity to perform daily living activities in healthy, as well as frail elderly (Bassey et al., 1992; Casas-Herrero et al., 2013; Reid and Fielding, 2012; Rech et al., 2014; Lopez et al., 2017); second, muscle power output and rate of force development decline faster than muscle strength in this population (Izquierdo et al., 1999a, 1999b); third, some studies and meta-analysis have shown that RT aimed to improve muscle power output (i.e., explosive RT) is more effective to improve functional abilities (i.e., sit-to-stand, walking ability, stairs climbing) than traditional RT (i.e., slower velocity of concentric muscle actions) (Bean et al., 2009; Miszko et al., 2003; Bottaro et al., 2007; Tschopp et al., 2011; Ramirez-Campillo et al., 2014; Steib et al., 2010); and fourth, explosive RT is a feasible exercise intervention even in institutionalized oldest old frail individuals (Cadore et al., 2013, 2014). It was reported that 12 weeks of multicomponent exercise training including explosive RT improved muscle power output (96–116%), strength (24–144%), muscle cross-sectional area, and muscle fat infiltration (4–8%), as well as functional outcomes, and dual task performance (7–58%) in frail institutionalized nonagenarians (Cadore et al., 2013). In another study, it was showed that 4 weeks of high-speed RT combined with walking, cognitive and balance exercises improved gait ability, balance, and muscle strength (15–30%), as well as reduced the incidence of falls in frail patients with dementia after long-term of physical restraint used in their nursing care (Cadore et al., 2014). Although in these studies in frail elderly populations the explosive RT was combined with walking and balance exercises, the RT was the main component of the intervention (Cadore et al., 2013, 2014).

The improvement in the neuromuscular function is even more important than muscle hypertrophy in elderly, and this goal should be focused when prescribing RT in this population. In this sense, explosive RT seems to be the most effective intervention in order to provide this benefit. It should be also mentioned, that the cardiovascular risk involved in this type of RT is even lower than the traditional type because heart rate, systolic blood pressure and cardiac output increase less during high-speed RT when compared with slow-speed RT (Lamotte et al., 2010; Miyamoto et al., 2017).

In line with our arguments, there are some aspects that should be taking into consideration when prescribing explosive RT in elderly individuals. Explosive RT should be performed with the concentric phase as fast as possible followed by a controlled, slower eccentric phase (Häkkinen et al., 1998, 2001; Izquierdo et al., 2001). The sets using explosive muscle actions can be performed alone (Bottaro et al., 2007; Cadore et al., 2013; Ramirez-Campillo et al., 2014), or combined with traditional RT (i.e., slower concentric muscle actions) (Häkkinen et al., 1998, 2001; Izquierdo et al., 2001) during the periodization. When performing explosive RT, sets until concentric failure should be avoided because fatigue may impair the muscle power output adaptations (Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010). The training intensity should range from 30 to 60% of 1RM load because maximal power output is maximized at these intensities (Izquierdo et al., 1999b). Although several studies have used standard weight machines for

power training (Häkkinen et al., 1998; Izquierdo et al., 2001; Bottaro et al., 2007; Cadore et al., 2013; Ramirez-Campillo et al., 2014), there are studies that used pneumatic machines for this type of RT (Fielding et al., 2002; Reid et al., 2015). A recent study has compared both types of machines during power training in older adults (Balachandran et al., 2017). These authors have shown similar improvements in muscle power output, physical performance battery (PPB), gallon jug transfer, muscle strength, timed-up-and-go or self-reported function, suggesting that both types of machines are effective when prescribing explosive RT in elderly (Balachandran et al., 2017). Finally, although RT is an effective exercise intervention to improve neuromuscular function and functional abilities, RT should be combined to any type of endurance exercise in order to improve cardiovascular function, because aging is associated with declines in both neuromuscular and cardiovascular functions (Cadore et al., 2012; Cadore and Izquierdo, 2013).

In summary, based on evidence provided by literature, although low volume resistance training may be an efficient intervention in order to improve muscle strength and hypertrophy, greater volume may optimize these gains in the long-term, especially in lower limbs. In addition, there is no evidence that performing repetitions until concentric failure may provide additional neuromuscular adaptations in elderly, and repetitions to failure promote greater increases in heart rate and blood pressure, which may increase the cardiovascular risk. Moreover, explosive resistance training must be prescribed in healthy and frail elderly individuals, at least in combination with traditional resistance training, because this type of training optimizes functional abilities gains, reduces incidence of falls, improves muscle strength and power output, and stimulates muscle hypertrophy. Finally, we appreciated the work of Fisher et al. (2017), and their effort to bring information to improve the resistance training prescription in elderly. Our purpose with this letter was to complement those information, as well as bring solid evidence that, in opposition to what they recommended, explosive resistance training should be performed by elderly individuals.

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